

Problem 6.2. Consider a random variable Y whose cumulative distribution function is given by

$$F_Y(y) = \begin{cases} 0, & \text{for } y < 0 \\ y^4, & \text{for } 0 \leq y < 1 \\ 1, & \text{for } 1 \leq y \end{cases}$$

Calculate the expectation of the random variable Y .

→ :

$$f_Y(y) = F'_Y(y) = \begin{cases} 4y^3 & \text{for } 0 \leq y < 1 \\ 0 & \text{otherwise} \end{cases} = 4y^3 \mathbb{1}_{[0,1)}(y)$$

$$\mathbb{E}[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy = \int_0^1 y \cdot 4y^3 dy = 4 \int_0^1 y^4 dy = \boxed{\frac{4}{5}}$$



The Normal Dist'n.

$$Y \sim N(\mu, \sigma^2)$$

$\left\{ \begin{array}{l} \mu \dots \text{mean} \\ \sigma \dots \text{standard deviation} \end{array} \right.$

The pdf: $f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$ for all $y \in \mathbb{R}$

The cdf. for all $y \in \mathbb{R}$:

$$F_Y(y) = \mathbb{P}[Y \leq y]$$

$$= \int_{-\infty}^y f_Y(u) du$$

$$= \int_{-\infty}^y \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(u-\mu)^2}{2\sigma^2}} du$$

NO ANALYTIC CDF!

Fact:

$$\frac{Y - \mu_Y}{\sigma_Y} \sim N(0, 1) \sim Z$$

$$Y = \mu_Y + \sigma_Y \cdot Z$$

$$F_Y(y) = \mathbb{P}[Y \leq y]$$

$$= \mathbb{P}\left[\frac{Y - \mu_Y}{\sigma_Y} \leq \frac{y - \mu_Y}{\sigma_Y}\right]$$

$$= \mathbb{P}\left[Z \leq \frac{y - \mu_Y}{\sigma_Y}\right]$$

$$= \Phi\left(\frac{y - \mu_Y}{\sigma_Y}\right)$$

$\Phi \dots$ cdf of $N(0, 1)$

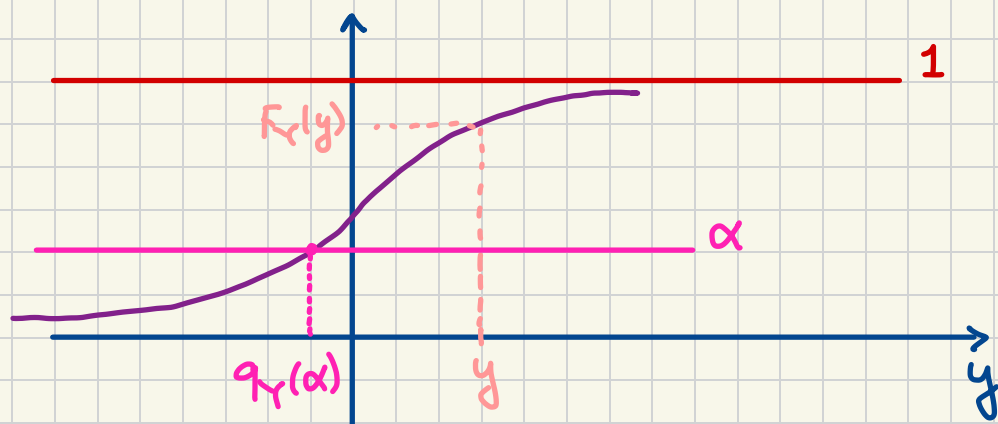
The Quantile Function.

Def'n. Let Y be a r.v. w/ cdf F_Y .
For $\alpha \in (0,1)$, the α -quantile of the dist'n of Y
is defined as the number $q_Y(\alpha) \in \mathbb{R}$
such that

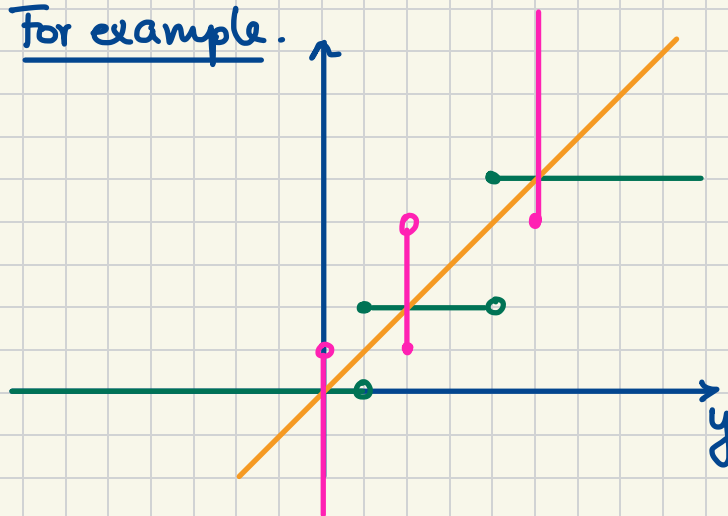
$$\mathbb{P}[Y \leq q_Y(\alpha)] = \alpha$$

$\Leftrightarrow$

$$F_Y(q_Y(\alpha)) = \alpha$$



For example.



$\nexists$ inverse $\ddot{}$

For the normal, this is not a problem!

The normal cdf is strictly increasing $\Rightarrow \exists$ an inverse

IF F_Y^{-1} exists, then

$$q_Y(\alpha) = F_Y^{-1}(\alpha)$$

M378K Introduction to Mathematical Statistics

Problem Set #7

Cumulative distribution functions: Named continuous distributions.

Problem 7.1. Source: Sample P exam, Problem #126.

A company's annual profit is normally distributed with mean 100 and variance 400. Then, we can express the probability that the company's profit in a year is at most 60, given that the profit in the year is positive in terms of the standard normal cumulative distribution function denoted by Φ as

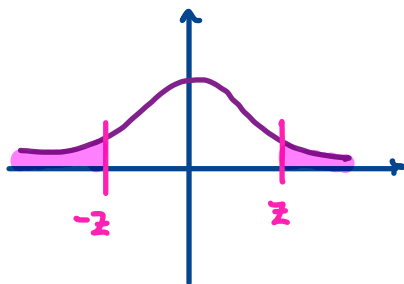
$$1 - \frac{\Phi(2)}{\Phi(5)}$$

True or false? Why?

→: $Y \sim N(\mu=100, \sigma^2=400)$

$$\begin{aligned} \mathbb{P}[Y \leq 60 \mid Y > 0] &= \frac{\mathbb{P}[0 < Y \leq 60]}{\mathbb{P}[Y > 0]} \\ &= \frac{\mathbb{P}\left[\frac{0-100}{20} < \frac{Y-100}{20} \leq \frac{60-100}{20}\right]}{\mathbb{P}\left[\frac{Y-100}{20} > \frac{0-100}{20}\right]} \end{aligned}$$

$$\begin{aligned} &= \frac{\mathbb{P}[-5 < Z \leq -2]}{\mathbb{P}[-5 < Z]} \\ &= \frac{\Phi(-2) - \Phi(-5)}{1 - \Phi(-5)} \\ &= \frac{1 - \Phi(2) - (1 - \Phi(5))}{\Phi(5)} \\ &= \frac{\Phi(5) - \Phi(2)}{\Phi(5)} \\ &= 1 - \frac{\Phi(2)}{\Phi(5)} \quad (\text{T}) \end{aligned}$$



$$\begin{aligned} \Phi(-z) &= \mathbb{P}[Z > z] \\ &= 1 - \mathbb{P}[Z \leq z] \\ &= 1 - \Phi(z) \end{aligned}$$

